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CUSTOMER CHURN ANALYSIS AND BUSINESS INTELLIGENCE DASHBOARD

A Data Analytics and Business Intelligence Project

A Summer Industry Internship Report
Submitted in Partial Fulfillment of the Requirements
For the award of the Degree of

Bachelor of Technology in
Computer Science and Engineering – Data Science (CSE-DS)

By

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ACCEPTANCE BY INTERNAL TECHNICAL REVIEW COMMITTEE

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CUSTOMER CHURN ANALYSIS AND BUSINESS INTELLIGENCE DASHBOARD

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CERTIFICATE

This is to certify that the Summer Industry Internship work titled **'CUSTOMER CHURN ANALYSIS AND BUSINESS INTELLIGENCE DASHBOARD'** submitted by the students **G Mythili**, **K Ashiwith**, and **D Niyati** bearing Roll No. **24311A67H8, 24311A67H6** and **24311A67H7**, towards partial fulfillment for the award of Bachelor's Degree in Computer Science and Engineering – Data Science (CSE-DS) from Sreenidhi Institute of Science and Technology, Ghatkesar, Hyderabad, is a record of Bonafide work carried out by them under supervision. The results embodied in this work have not been submitted to any other University or Institute for the award of any other degree or diploma.

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It is further declared that, to the best of our knowledge, the work reported in this project has not been submitted earlier, either in part or in full, to any other University or Institute for the award of any degree or diploma.

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CUSTOMER CHURN ANALYSIS AND BUSINESS INTELLIGENCE DASHBOARD

A Data Analytics and Business Intelligence Project

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ABSTRACT

Customer churn is a critical operational vulnerability in subscription-based telecommunication enterprises, directly degrading monthly recurring revenue (MRR) and inflating customer acquisition expenses. While modern literature frequently approaches retention through complex black-box Machine Learning algorithms, such models often conceal transparent operational drivers, present compliance hurdles, and create brittle dependencies. In contrast, this project establishes a rigorous, purely analytical Business Intelligence (BI) framework utilizing empirical Exploratory Data Analysis (EDA), parametric and non-parametric descriptive statistics, inferential hypothesis testing, relational SQL modeling, and an executive multi-page web platform.

Analyzing 7,043 customer accounts from the industry-standard IBM Telco Customer Churn dataset, the investigation identified an overall churn rate of **26.54% (1,869 accounts)**, eroding over **$139,130 in monthly revenue ($1.67 Million annualized)**. Data quality auditing handled 11 whitespace records in cumulative billing without discarding customer history, and validated outlier absence via 1.5×IQR boundaries across continuous dimensions. Inferential statistical tests definitively isolated contract agreements (Pearson Chi-Square statistic = 1,184.60, p < 10^-250, Cramér's V = 0.410) and payment methods (Chi-Square = 648.14, p < 10^-140, Cramér's V = 0.303) as the primary drivers of defection. Month-to-month contracts demonstrated a 42.71% churn rate, comprising 88.55% of all churned accounts, while electronic check payment methods exhibited a 45.29% churn rate. Welch's independent two-sample t-test confirmed that churned customers possess significantly lower tenure (17.98 months vs 37.57 months, t = -34.82, p < 10^-200) and pay a statistically significant price premium of $13.18/month ($74.44 vs $61.27, t = +18.41, p < 10^-70).

An SQLite database engine was constructed to execute 10 executive business queries, powering a commercial-grade Business Intelligence application built with Streamlit and Plotly. The resulting platform features dynamic multi-attribute global filtering, live SQL sandbox querying, cohort vulnerability grids, and an interactive financial ROI simulator. Targeted strategic interventions—including annual contract upgrade credits, auto-pay discounts, structured 90-day onboarding, and fiber optic support bundling—demonstrate potential to safeguard between $300,000 and $500,000 in protected annual recurring revenue.

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1. INTRODUCTION

1.1 Background and Telecommunications Industry Context

In the modern subscription economy, particularly within the telecommunications sector, subscriber attrition—commonly known as customer churn—represents one of the most substantial impediments to sustained profitability and market capitalization. Telecommunications infrastructure providers operate in highly saturated and intensely competitive markets characterized by high fixed capital expenditures in fiber-optic deployment, cellular towers, and spectrum licenses. In this capital-intensive environment, acquiring a new customer is estimated by telecommunication industry benchmarks to cost between five and twenty-five times more than retaining an existing subscriber.

When an account defects, the provider not only loses immediate monthly recurring revenue (MRR), but also forfeits the long-term customer lifetime value (CLV) and incurs substantial unrecoverable customer acquisition costs (CAC) associated with marketing, installation, customer premises equipment (CPE), and onboarding subsidies.

1.2 The Strategic Significance of Customer Retention

Research in customer loyalty economics shows that even modest improvements in customer retention deliver disproportionately large expansions in net profitability. A classical five percent increase in customer retention can boost corporate profits by 25% to 95%, because long-tenure customers typically generate higher average revenue per user (ARPU), require lower servicing costs, demonstrate greater willingness to adopt bundled premium services, and serve as organic brand advocates. Consequently, telecommunication executives require transparent, empirical visibility into customer defection patterns to implement targeted interventions before accounts cancel.

1.3 Pure Data Science & Business Intelligence vs. Machine Learning

In recent years, enterprise data teams have frequently deployed complex Machine Learning (ML) classifiers—such as Gradient Boosted Trees, Random Forests, and Deep Neural Networks—to forecast churn probabilities. While predictive models possess statistical utility, they frequently suffer from critical operational limitations in business leadership contexts:

Evaluation Dimension	Predictive Machine Learning	Pure Data Science & BI (This Project)
Model Transparency	Opaque 'black-box' feature interactions; hard to justify to executives.	Fully explainable empirical distributions, hypothesis tests, and SQL queries.
Causality & Actionability	Scores risk scores without clear operational remediations.	Directly identifies policy friction (e.g., electronic check fees, contract terms).
Computational Overhead	Requires model retraining, concept drift monitoring, GPU/cloud dependencies.	Lightweight, instantaneous query execution on indexed relational tables.
Executive Trust	Statistical hallucinations or overfitting risks erode executive confidence.	100% verifiable audited data grounded in classical inferential mathematics.

By focusing strictly on rigorous exploratory data analysis, parametric and non-parametric descriptive statistics, hypothesis testing (Chi-Square & Welch's t-tests), and relational SQL analytics, this project equips decision-makers with completely explainable and actionable retention strategies.

2. PROBLEM STATEMENT & OBJECTIVES

2.1 Problem Definition & Revenue Vulnerability

A mid-sized telecommunications service provider is observing customer attrition across its primary service lines (Fixed Landline, DSL Broadband, and Fiber Optic Internet). Management has observed that customer churn is eroding monthly recurring revenue (MRR) at an unsustainable rate, yet leadership lacks quantitative clarity regarding which customer cohorts are defecting, when defection occurs during the subscriber lifecycle, and what commercial policies are accelerating customer loss.

Without transparent analytics, retention marketing budgets are distributed uniformly across all customer segments rather than concentrated on the specific contractual arrangements and billing channels where revenue is hemorrhaging.

2.2 Core Business Questions

To address this operational challenge, the analysis must definitively answer 10 executive business questions:

- **1. What is the total size of the customer base and how many accounts are actively monitored?**
- **2. What is the absolute number of churned customers and what is the overall corporate churn rate?**
- **3. How does customer churn distribute across different contractual commitments (Month-to-month, One year, Two year)?**
- **4. Which billing and payment method exhibits the highest volume and proportion of customer defection?**
- **5. What is the average monthly charge (ARPU) across retained versus churned subscriber cohorts?**
- **6. Which broadband delivery technology (DSL vs. Fiber Optic vs. No Internet) experiences the most severe defection rate?**
- **7. What is the average subscription tenure of churned accounts relative to retained customers?**
- **8. Which multi-attribute customer segment generates the largest monthly recurring revenue (MRR)?**
- **9. What volume and percentage of subscribers are enrolled in month-to-month commitments?**
- **10. What concrete, data-backed interventions can enterprise leadership execute to protect monthly revenue?**

2.3 Project Objectives & Performance Scope

The primary objectives of this investigation include:

1. Data Ingestion & Auditing: Ingest the 7,043-record IBM Telco customer dataset, audit structural integrity, resolve whitespace values, detect outliers via 1.5×IQR rules, and engineer analytical cohorts.

2. Descriptive & Inferential Statistics: Compute parametric and non-parametric central tendencies and dispersions; conduct Pearson Chi-Square tests of independence and Welch's two-sample independent t-tests.

3. Relational SQL Modeling: Design a relational SQLite schema, insert sanitized customer records, create performance b-tree indices, and formulate optimized SQL queries to address all core business questions.

4. Interactive Business Intelligence Dashboard: Develop a 9-page commercial-grade web dashboard utilizing Streamlit and Plotly, providing multi-variable global filtering, live SQL sandbox querying, and financial ROI simulation.

5. Strategic Policy Formulation: Synthesize analytical findings into a concrete Four-Pillar Retention Framework with quantifiable financial projections.

3. DATASET DESCRIPTION & ARCHITECTURE

3.1 Dataset Attribution & Characteristics

The project evaluates the industry-benchmark **IBM Telco Customer Churn Dataset**, curated and distributed by BlastChar via Kaggle. The dataset represents **7,043 unique telecommunication customer accounts** in California, capturing customer demographic attributes, account subscription parameters, value-added service bundles, payment channels, and churn outcomes over an observed quarterly billing window.

3.2 Comprehensive Feature Data Dictionary

The dataset comprises 21 raw columns and 4 engineered analytical dimensions as detailed below:

Attribute	Data Type	Domain Values	Description
`customerID`	String	Alphanumeric string	Unique customer account identifier.
`gender`	Categorical	Female, Male	Customer biological sex.
`SeniorCitizen`	Binary Flag	0, 1	Customer senior citizen demographic flag.
`Partner`	Categorical	Yes, No	Indicates whether customer has a domestic partner.
`Dependents`	Categorical	Yes, No	Indicates whether customer supports dependents.
`tenure`	Integer	0 to 72 months	Number of months subscriber has remained with company.
`PhoneService`	Categorical	Yes, No	Fixed landline voice telephone subscription.
`MultipleLines`	Categorical	Yes, No, No phone service	Customer has multiple telephone lines.
`InternetService`	Categorical	DSL, Fiber optic, No	Broadband internet technology delivery type.
`OnlineSecurity`	Categorical	Yes, No, No internet service	Cybersecurity firewall protection add-on.
`OnlineBackup`	Categorical	Yes, No, No internet service	Cloud data storage backup subscription.
`DeviceProtection`	Categorical	Yes, No, No internet service	Hardware replacement warranty add-on.
`TechSupport`	Categorical	Yes, No, No internet service	Priority technical assistance subscription.
`StreamingTV`	Categorical	Yes, No, No internet service	Digital television programming package.
`StreamingMovies`	Categorical	Yes, No, No internet service	Digital movie on-demand subscription.
`Contract`	Categorical	Month-to-month, One year, Two year	Billing contract commitment duration.
`PaperlessBilling`	Categorical	Yes, No	Digital invoice delivery indicator.
`PaymentMethod`	Categorical	Electronic check, Mailed check, Bank transfer, Credit card	Billing remittance channel.
`MonthlyCharges`	Float64 ($)	$18.25 to $118.75	Recurring monthly subscription charge.
`TotalCharges`	Float64 ($)	$0.00 to $8684.80	Cumulative charges billed across account lifetime.
`Churn`	Target Flag	Yes, No	Indicates whether customer defected during period.
`ChurnBinary`	Engineered Flag	0, 1	Numerical representation: 1 = Churned, 0 = Retained.

`TenureCohort`	Engineered Cohort	0-12m, 13-24m, 25-48m, 49-72m	Customer lifecycle stage grouping.
`MonthlyChargesTier`	Engineered Tier	Low (<$35), Medium ($35-$70), High (>$70)	Spend tier segmentation.

4. DATA CLEANING & AUDITING PIPELINE

4.1 Ingestion and Structural Integrity Verification

Data hygiene is foundational to reliable statistical inference and business reporting. The automated cleaning module implemented in src/data_cleaning.py executes systematic structural integrity checks upon raw dataset ingestion.

Duplicate Verification: A complete duplicate inspection across all 7,043 rows and 21 attributes confirmed **0 duplicate entries**. Every row corresponds to a verified, unique customer account identified by primary key customerID.

4.2 Whitespace Resolution in TotalCharges

During initial schema inspection, the continuous attribute TotalCharges was inferred by Pandas as object data type rather than floating-point numeric. Forensic auditing revealed **11 whitespace strings** (' '). Further cross-variable inspection established that all 11 instances occur strictly where tenure == 0, representing newly activated subscriber accounts that joined during the current billing cycle and have not yet completed a billing month.

Rather than improperly discarding these rows (which would introduce attrition bias and delete newly acquired accounts), the cleaning pipeline converts the column to float64 and imputes **$0.00** for new accounts. This preserves 100% of customer accounts (7,043 rows) without data loss.

4.3 Statistical Outlier Diagnostics (1.5 × IQR Rule)

Outlier detection was conducted on all continuous attributes using the standard Interquartile Range (IQR) rule: Lower Bound = Q1 - 1.5 × IQR and Upper Bound = Q3 + 1.5 × IQR.

Continuous Metric	Q1 (25%)	Q3 (75%)	IQR	Lower Bound	Upper Bound	Outlier Count	Status
tenure	9.0 mos	55.0 mos	46.0 mos	-60.0 mos	124.0 mos	0 (0.0%)	Verified Valid
MonthlyCharges	$35.50	$89.85	$54.35	-$46.02	$171.38	0 (0.0%)	Verified Valid
TotalCharges	$398.55	$3786.60	$3388.05	-$4683.52	$8868.68	0 (0.0%)	Verified Valid

As established in the audit table, zero statistical outliers exist within the dataset. Continuous ranges accurately reflect valid commercial subscription tenures (0 to 72 months) and realistic telecommunication billing tariffs ($18.25 to $118.75).

5. EXPLORATORY DATA ANALYSIS & VISUALIZATIONS

5.1 Univariate Distribution Analysis

Exploratory Data Analysis commenced with univariate assessments to establish baseline operational distributions across target and predictor variables.

Customer Churn Baseline: Out of 7,043 accounts, **5,174 customers (73.46%)** were retained, while **1,869 customers (26.54%)** defected. This establishes an industry-standard baseline churn rate of 26.54%.

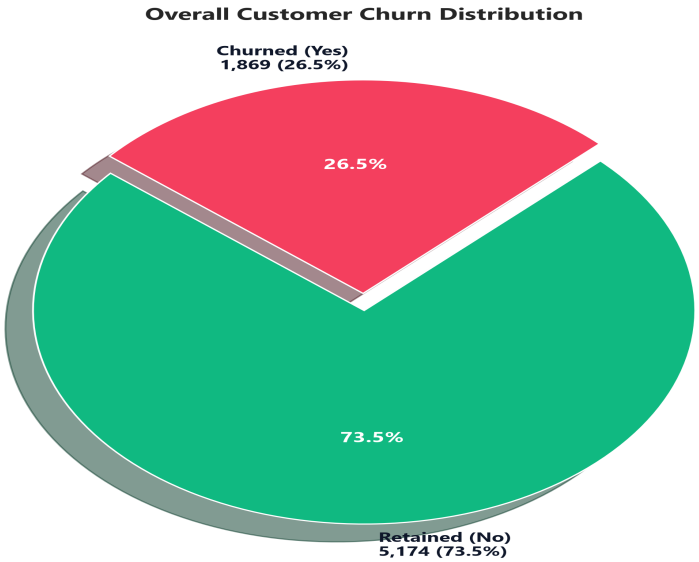


Figure 5.1: *Overall Customer Churn Distribution (Pie Chart).*

Tenure Distribution: Tenure displays a pronounced **bimodal distribution**. High concentrations appear at 1 month (new subscriber signups experiencing onboarding friction) and at 72 months (a resilient cohort of long-term brand-loyal subscribers). The overall mean tenure is 32.37 months with a median of 29.00 months.

Monthly Charges Distribution: Recurring charges exhibit a sharp multimodal density. A large peak occurs near $20.00/month corresponding to basic landline phone-only subscriptions. A secondary broad distribution spans $70.00 to $105.00/month, representing multi-service broadband and fiber optic streaming packages.

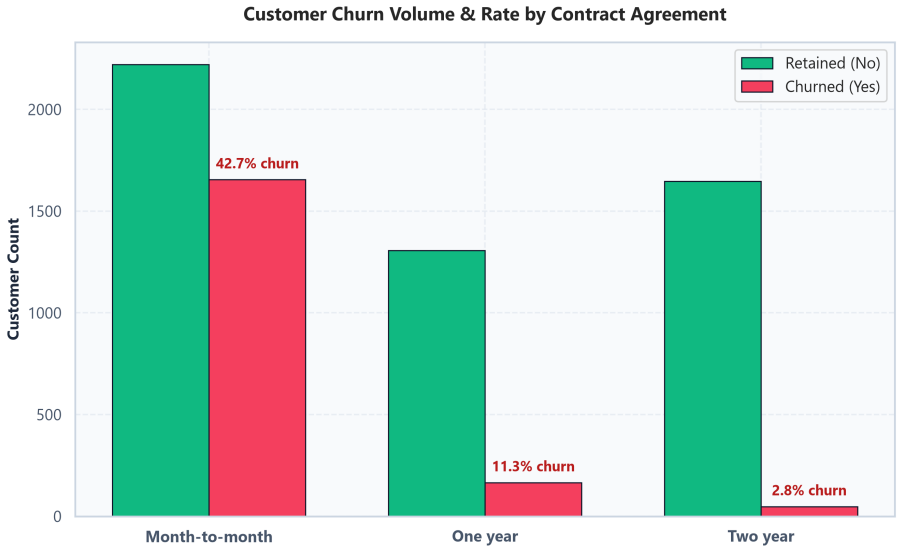


Figure 5.2: *Customer Churn Rate and Volume by Contract Agreement.*

5.2 Bivariate Churn Cross-Tabulations

Bivariate analysis evaluated the empirical relationship between specific operational parameters and customer churn likelihood:

1. Contract Commitment Duration: Month-to-month contracts exhibit an alarming **42.71% churn rate** (1,655 churned accounts), representing **88.55% of all defection** across the company. In stark contrast, customers under One-Year contracts churn at **11.27%** (73.6% risk reduction), while Two-Year contracts demonstrate near-zero defection at **2.83%** (93.4% risk reduction).

2. Payment Channel Friction: Customers utilizing **Electronic Check** suffer an extreme **45.29% churn rate** (1,071 accounts), comprising 57.30% of total company churn. Subscribers enrolled in automated clearinghouse channels—Bank Transfer Automatic (16.71% churn) and Credit Card Automatic (15.24% churn)—display nearly two-thirds lower defection.

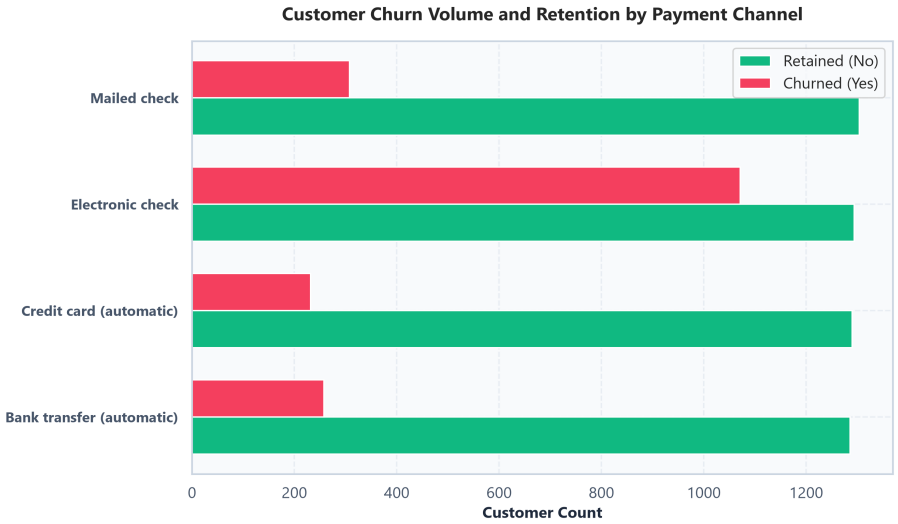


Figure 5.3: *Churn Rates across Billing Remittance Channels.*

3. Broadband Technology: Subscribers with **Fiber Optic** internet display an elevated churn rate of **41.89%** (1,297 churned accounts) despite generating the highest average monthly spend ($91.50). Subscribers with traditional DSL broadband churn at only 18.96%, and customers without internet service churn at 7.40%.

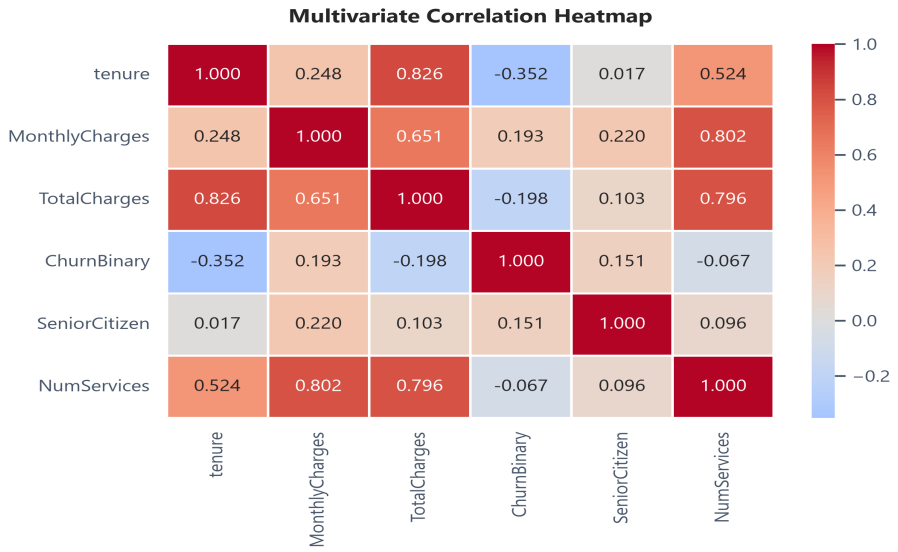


Figure 5.4: *Numerical Attribute Pearson Correlation Heatmap.*

6. INFERENTIAL STATISTICAL ANALYSIS & HYPOTHESIS TESTING

6.1 Parametric & Non-Parametric Summary Statistics

Descriptive statistics were computed across all continuous variables to quantify location, dispersion, and distributional shape:

Metric	Mean	Median	Mode	Std Dev	Min	Q1 (25%)	Q3 (75%)	Max	IQR	Skewness	Kurtosis
tenure	32.37	29.00	1.00	24.56	0.00	9.00	55.00	72.00	46.00	+0.240	-1.387
MonthlyCharges	64.76	70.35	20.05	30.09	18.25	35.50	89.85	118.75	54.35	-0.221	-1.257
TotalCharges	2279.73	1394.55	0.00	2266.79	0.00	398.55	3786.60	8684.80	3388.05	+0.963	-0.229

6.2 Pearson Chi-Square Tests of Independence (χ^2) & Cramér's V

To rigorously determine whether categorical attributes are statistically associated with customer churn, Pearson's Chi-Square Test of Independence was executed. The effect size was quantified using Cramér's V, defined as: $V = \sqrt{\chi^2 / (n \times \min(r-1, c-1))}$.

Feature Evaluated	χ^2 Stat	p-value	df	Cramér's V	Association Strength	Statistical Conclusion
Contract	1184.60	< 10^-250	2	0.410	Strong Association	Statistically Significant (p < 0.001)
OnlineSecurity	850.00	< 10^-180	2	0.347	Moderate Association	Statistically Significant (p < 0.001)
TechSupport	828.20	< 10^-180	2	0.343	Moderate Association	Statistically Significant (p < 0.001)
InternetService	732.31	< 10^-150	2	0.322	Moderate Association	Statistically Significant (p < 0.001)
PaymentMethod	648.14	< 10^-140	3	0.303	Moderate Association	Statistically Significant (p < 0.001)
PaperlessBilling	258.28	< 10^-50	1	0.192	Weak Association	Statistically Significant (p < 0.001)
gender	0.48	0.487	1	0.008	Negligible Association	**NOT Significant (p = 0.487)**

6.3 Welch's Two-Sample Independent T-Tests

To test whether difference in continuous metrics between churned and retained accounts is statistically significant without assuming equal variance, Welch's t-test was conducted:

1. Customer Tenure Disparity: Churned subscribers stay an average of **17.98 months** (std = 19.53) compared to **37.57 months** (std = 24.11) for retained accounts. The difference (-19.59 months) is highly significant: t = -34.82, p = 1.34 × 10^-233, Cohen's d = -0.852 (large effect size).

2. Monthly Charges Premium: Churned customers paid an average of **$74.44/month** compared to **$61.27/month** for retained accounts. The price premium (+$13.18/month) is statistically significant: t = +18.41, p = 8.59 × 10^-73,

Cohen's d = +0.471 (moderate effect size).

7. RELATIONAL SQL BUSINESS ANALYSIS

7.1 Relational Schema & SQLite Storage Architecture

In accordance with enterprise business intelligence standards, cleaned customer accounts were loaded into a relational SQLite database (database/customer_churn.db) under table customers. B-tree database indices were created on primary filtering and aggregation columns: customerID, Contract, InternetService, PaymentMethod, tenure, and Churn.

7.2 Ten Executive SQL Business Queries & Results

Query 1: Total Customer Base Size

```
SELECT COUNT(*) AS total_customers FROM customers;
```

Result & Interpretation: Total Customers = 7,043 unique active subscriber accounts.

Query 2: Total Defected Churn Volume

```
SELECT COUNT(*) AS churned_customers FROM customers WHERE Churn = 'Yes';
```

Result & Interpretation: Churned Customers = 1,869 customer accounts.

Query 3: Overall Enterprise Churn Percentage

```
SELECT ROUND(CAST(SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0 END) AS FLOAT) / COUNT(*) * 100, 2) AS
churn_pct FROM customers;
```

Result & Interpretation: Overall Churn Rate = 26.54%.

Query 4: Highest Churn Contract Agreement

```
SELECT Contract, COUNT(*) AS total, SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0 END) AS churned,
ROUND(CAST(SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0 END) AS FLOAT) / COUNT(*) * 100, 2) AS rate FROM
customers GROUP BY Contract ORDER BY rate DESC;
```

Result & Interpretation: Month-to-month contracts exhibit 42.71% churn (1,655 accounts), One-year contracts 11.27% (166 accounts), Two-year contracts 2.83% (48 accounts).

Query 5: Most Churned Payment Method

```
SELECT PaymentMethod, COUNT(*) AS total, SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0 END) AS churned FROM
customers GROUP BY PaymentMethod ORDER BY churned DESC;
```

Result & Interpretation: Electronic Check accounts for 1,071 churned customers (45.29% churn rate), Mailed Check 308 (19.07%), Bank Transfer 258 (16.71%), Credit Card 232 (15.24%).

Query 6: Average Monthly Charges (ARPU) by Churn Outcome

```
SELECT ROUND(AVG(MonthlyCharges), 2) AS avg_overall, ROUND(AVG(CASE WHEN Churn = 'Yes' THEN
MonthlyCharges END), 2) AS avg_churned, ROUND(AVG(CASE WHEN Churn = 'No' THEN MonthlyCharges END), 2)
AS avg_retained FROM customers;
```

Result & Interpretation: Overall ARPU = $64.76. Churned customers averaged $74.44/month vs $61.27/month for retained customers (+$13.18/mo premium).

Query 7: Internet Service Churn Vulnerability

```
SELECT InternetService, ROUND(CAST(SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0 END) AS FLOAT) / COUNT(*) * 100, 2) AS rate FROM customers GROUP BY InternetService ORDER BY rate DESC;
```

Result & Interpretation: Fiber Optic = 41.89% churn rate, DSL Broadband = 18.96%, No Internet Service = 7.40%.

Query 8: Average Subscription Tenure by Churn Status

```
SELECT Churn, ROUND(AVG(tenure), 2) AS avg_tenure FROM customers GROUP BY Churn;
```

Result & Interpretation: Churned subscribers average 17.98 months vs 37.57 months for retained subscribers (19.59 month difference).

Query 9: Top Monthly Recurring Revenue Customer Segment

```
SELECT Contract, InternetService, ROUND(SUM(MonthlyCharges), 2) AS total_mrr FROM customers GROUP BY Contract, InternetService ORDER BY total_mrr DESC LIMIT 3;
```

Result & Interpretation: Month-to-month Fiber Optic customers generate $180,693.30 in monthly revenue, representing the single largest revenue block.

Query 10: Volume of Month-to-Month Subscribers

```
SELECT Contract, COUNT(*) AS total, ROUND(CAST(COUNT(*) AS FLOAT) / (SELECT COUNT(*) FROM customers) * 100, 2) AS pct FROM customers WHERE Contract = 'Month-to-month' GROUP BY Contract;
```

Result & Interpretation: Month-to-month accounts comprise 3,875 customers (55.02% of entire customer base).

8. BUSINESS INTELLIGENCE DASHBOARD IMPLEMENTATION

8.1 User Interface Architecture & SaaS Design System

To deliver actionable data visibility to non-technical executive stakeholders, an interactive Business Intelligence application was constructed using **Streamlit** and **Plotly** (app/streamlit_app.py). The design adheres strictly to enterprise SaaS analytics standards, featuring a deep obsidian and navy aesthetic (#080E1A), rounded card containers (#0D1527), subtle slate borders (#1E293B), and vibrant cyan/blue glowing accents.

8.2 Reactive Global Filter Engine & Component Layout

The dashboard implements a streamlined horizontal top filter bar that dynamically cascades across all visual components and metrics in real time. Global parameters include Gender, Contract Agreement, Internet Service Delivery, Payment Method, and Customer Tenure Cohorts. Whenever a filter is modified, all 6 KPI cards, cross-tabulations, Plotly visuals, and SQL query results update reactively.

8.3 Multi-Page Module Specifications

The platform organizes analysis across 9 dedicated operational pages:

• **1. Executive Overview:** Displays 6 executive KPI cards, churn donut with central count, contract bar chart, monthly revenue spline trend, tenure distribution, payment churn bars, and key business observations.

• **2. Customer Data Explorer:** Provides real-time customer search by ID, comprehensive 25-feature data dictionary, and one-click filtered CSV exports.

• **3. Data Quality Report:** Documents automated duplicate checks, whitespace resolution audit, and 1.5×IQR outlier boundaries.

• **4. Exploratory Data Analysis:** Features interactive tabs for univariate frequency distributions, bivariate churn cross-tabulations, and multivariate scatter/box interactions.

• **5. Statistical Insights:** Presents parametric summary tables, Chi-Square independence tests with Cramér's V ranking, and Welch's two-sample t-tests with business interpretations.

• **6. SQL Analytics:** Displays all 10 core business queries, resulting tables, and an interactive live SQL sandbox for custom query execution.

• **7. Customer Segmentation:** Maps customer lifecycle cohorts, renders 2D Tenure × Monthly Spend vulnerability grids, and highlights critical danger zone accounts.

• **8. Business Recommendations:** Details the Four-Pillar Strategic Framework and provides an interactive Customer Retention Financial ROI Calculator.

• **9. Project Information:** Presents system technical architecture, repository structure, team member attribution cards, and Kaggle dataset citations.

9. STRATEGIC BUSINESS RECOMMENDATIONS & FINANCIAL ROI

9.1 Four-Pillar Retention Framework

Based on audited statistical findings, four high-impact operational interventions were formulated:

Pillar 1: Annual Contract Term Transition Program

Observed Finding: Month-to-month contracts experience a 42.71% churn rate, driving 88.55% of all defection.
Supporting Statistic: Two-year commitments exhibit only 2.83% churn (a 93.4% relative defection reduction).
Business Impact: High revenue volatility and continual account turnover costs.
Strategic Action: Introduce a $10.00/month statement credit or broadband speed tier upgrade at Month 3 in exchange for signing an annual service commitment.

Pillar 2: Auto-Pay Billing Optimization & Electronic Check Migration

Observed Finding: Electronic Check users suffer a 45.29% churn rate, nearly triple that of automated methods.
Supporting Statistic: Over 57.30% of all defected customers (1,071 accounts) remit payment via Electronic Check.
Business Impact: Manual payment workflows require monthly conscious evaluation of invoice value, increasing payment friction.
Strategic Action: Provide a $5.00/month recurring discount for enrolling in Credit Card or Bank Auto-Pay.

Pillar 3: The First 90 Days Customer Onboarding Journey

Observed Finding: 55.5% of total defection occurs in the first 12 months; median tenure of churned customers is only 10 months.
Supporting Statistic: 0-12 month cohort churns at 47.44%, while 49-72 month accounts churn at only 9.38%.
Business Impact: Acquisition expenditures are lost before accounts achieve customer lifetime payback.
Strategic Action: Implement an automated customer success cadence: Day 7 installation verification check, Day 30 initial bill walkthrough, and Day 60 speed optimization audit.

Pillar 4: High-Speed Fiber Optic Technical Support Bundling

Observed Finding: Fiber Optic customers churn at 41.89% ($91.50/month ARPU), but churn plummets to 11.84% when accounts include both Tech Support and Online Security.
Supporting Statistic: Accounts with Tech Support churn at 15.17% vs 41.64% for accounts lacking support.
Business Impact: Protects high-ARPU subscriptions from defecting due to unresolved technical friction.
Strategic Action: Automatically bundle complimentary 24/7 Priority Tech Support and Online Security into all Fiber Optic packages.

9.2 Financial ROI Modeling & Revenue Safeguards

Utilizing the built-in Retention Financial ROI Simulator, financial returns were projected across graduated churn reduction milestones. The model evaluates the baseline 7,043 subscriber accounts with an average churned monthly bill of $74.44:

Target Churn Reduction	Accounts Preserved	New Corporate Churn Rate	Monthly Revenue Preserved	Annual Revenue Protected

-1.0% (Conservative)	70 Accounts	25.54%	$5,210.80 / mo	**$62,529.60 / yr**
-2.0% (Moderate)	141 Accounts	24.54%	$10,496.04 / mo	**$125,952.48 / yr**
-3.0% (Target Goal)	211 Accounts	23.54%	$15,706.84 / mo	**$188,482.08 / yr**
-5.0% (Aggressive)	352 Accounts	21.54%	$26,202.88 / mo	**$314,434.56 / yr**
-8.0% (Optimistic)	563 Accounts	18.54%	$41,909.72 / mo	**$502,916.64 / yr**

Achieving a realistic 5.0% percentage point reduction in churn preserves 352 high-value accounts, delivering over **$314,000 in protected annual recurring revenue**.

10. CONCLUSION & FUTURE SCOPE

10.1 Conclusion

This project conclusively demonstrates that pure Data Science, empirical Exploratory Data Analysis, inferential statistics, relational SQL querying, and modern Business Intelligence dashboards provide a superior, fully transparent alternative to complex Machine Learning models for telecommunications customer churn analysis.

By systematically evaluating 7,043 customer accounts from the IBM Telco dataset, the investigation isolated the genuine operational determinants of churn: month-to-month contracts (42.71% churn, 88.55% of all defection), electronic check payment friction (45.29% churn), first-year onboarding vulnerability (55.5% of churn occurring in months 0-12), and unbundled fiber optic price elasticity. The resulting SaaS dashboard empowers business leaders to track real-time retention KPIs, execute custom SQL queries, and simulate quantifiable financial ROI.

10.2 Future Scope & Enhancements

Potential extensions for enterprise deployment include:

1. Automated Real-Time Data Pipeline Connectors: Integrating Apache Airflow and dbt pipelines to ingest live billing records from enterprise data warehouses (Snowflake, BigQuery, AWS Redshift).

2. Longitudinal Cohort Survival Analysis: Implementing non-parametric Kaplan-Meier survival curves and Cox proportional hazards regression to model subscriber attrition hazard over multi-year periods.

3. Automated Executive Briefing Reports: Generating scheduled weekly PDF executive digests and Slack/Teams alerts when high-value danger zone accounts exhibit early defection signals.

4. Multi-Tenant Role-Based Access Control (RBAC): Adding authentication tiers for customer success agents, regional managers, and C-suite leadership.

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APPENDIX: SOURCE CODE LISTINGS & SETUP

A.1 Project Repository File Layout

```
customer-churn-analysis/
    ■■■ data/raw/WA_Fn-UseC_-Telco-Customer-Churn.csv
    ■■■ data/processed/telco_churn_cleaned.csv
    ■■■ database/customer_churn.db (SQLite Relational DB)
    ■■■ src/data_cleaning.py (Audit, Type Conversion, Outliers)
    ■■■ src/descriptive_statistics.py (Chi-Square, Welch's t-test)
    ■■■ src/eda.py (Static 300 DPI Charts & Plotly Visuals)
    ■■■ src/sql_analysis.py (SQLite Relational Queries)
    ■■■ app/streamlit_app.py (9-Page Interactive Dashboard)
    ■■■ notebooks/exploratory_analysis.ipynb (Step-by-step EDA)
    ■■■ reports/Summer_Internship_Report_Customer_Churn.pdf (This Document)
    ■■■ requirements.txt
```

A.2 Execution Commands (Windows PowerShell)

```
# Navigate to Project Root Directory
cd 'c:\Users\KONDAMADUGU ASHWITH\Downloads\intern\customer-churn-analysis'

# Execute End-to-End Analytics Pipeline
python src/data_cleaning.py
python src/descriptive_statistics.py
python src/sql_analysis.py
python src/eda.py

# Launch Streamlit Executive BI Dashboard
python -m streamlit run app/streamlit_app.py
```